10. Acceptance-Rejection 2

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1. Beta $(1,1)$, the uniform distribution
2. Beta $(1/2, 1/2)$, the arcsin distribution
3. Beta $(3/2, 3/2)$, Wigner's semi-circle distribution after shifting and scaling.

$$F(x) = \frac{1}{4R^3}(2R - x)(R + x)^2$$

$$f(x) = \frac{3}{4R^3}(R^2 - x^2)$$

Recall from last time,

- Want to sample from $f(x)$
- We choose an $h(x)$ distribution to sample from instead.
- Write $f(x) = c\, g(x)\, h(x) \iff \dfrac{f(x)}{h(x)} = c\, g(x)$

Start Here

Find these

$c \geq 1$ $\quad \in [0,1]$ $\quad$ pdf

Algorithm

Require, $g(x)$ and $h(x)$

Repeat

- Sample / Generate Z from $h(z)$
- Sample / Generate U from $\text{Unif}[0,1]$

Independent

If $U \leq g(Z)$ (Accept)

Stop and return $X = Z$

Else Repeat the above (Reject)

Output $X \sim f(x)$

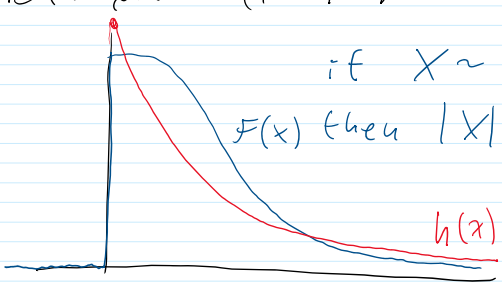
Question: How long until we get an acceptance and stop the algorithm?

Given $f(x)$ and $h(x)$, what is $P(\text{Accept})$?

$= P(U \leq g(Z)) = 1/c$

Comes right out of the proof from last time.

Example: (Half Normal Dist)



if $X \sim N(0,1)$

$f(x)$ then $|X| \sim$ Half Normal

$h(x) = e^{-x}$

$f(x)$ = Dist we want

c = Real # ≥ 1

$g(x)$ = returns $[0,1]$ ← Not a dist.

$h(x)$ = Dist we sample from

Warning: we don't want

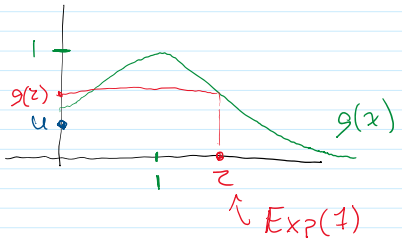
$f(x) = \sqrt{2}\, e^{-x^2/2}$

$$f(x) = \sqrt{\frac{2}{\pi}}\, e^{-x^2/2} \quad \text{for } x \geq 0$$

$$h(x) = e^{-x} \quad \text{for } x \geq 0$$

Warning: we don't want $\frac{f(x)}{h(x)} \to \infty$

$$f(x) = \underbrace{\sqrt{\frac{2e}{\pi}}}_{c}\, \underbrace{e^{-(x-1)^2/2}}_{g(x)}\, \underbrace{e^{-x}}_{h(x)} \qquad g(x) = e^{-(x-1)^2/2}$$



Algorithm say Accept if $u \leq g(z)$

$$P(\text{Accept}) = \frac{1}{c} = \sqrt{\frac{\pi}{2e}}$$

Reason: Bayes Formula

going to be $f(\cdot)$

$$h(z \mid u \leq g(z)) = \frac{\int_0^{g(z)} f_{uz}(u,z)\,du}{P(u \leq g(z))}$$

$g(z)h(z)$

$\frac{1}{c}$

← Normalizing so that the entire expression integrates to 1.

$$P(A \mid B) = \frac{P(A,B)}{P(B)}$$

$$\frac{f(x)}{h(x)} = \frac{\sqrt{\frac{2}{\pi}}\, e^{-x^2/2}}{e^{-x}} = \underbrace{\sqrt{\frac{2}{\pi}}}_{\leq 1}\, \underbrace{e^{-x^2/2 + x}}_{\text{Not in } [0,1]} = \sqrt{\frac{2}{\pi}}\, \exp\left[-\frac{(x-1)^2}{2} + \frac{1}{2}\right]$$

$$= \underbrace{\sqrt{\frac{2e}{\pi}}}_{\geq 1}\, \underbrace{\exp\left[-\frac{(x-1)^2}{2}\right]}_{\in [0,1]}$$

$P(\text{Accept}) = 1/c$ ← geometric$(1/c)$

Expected Value of Geometric

Average # of iterations until an acceptance $= c$

Take Away: Pick an $h(x)$ to make c as close to 1 as possible!